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Wave Optics

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	40 questions
Answer key	at the end

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MCQ (Single Correct Answer) 40 questions

Q1. NEET 2026

In Young's double slit experiment, using monochromatic light of wavelength λ, the intensity of light at a point on the screen where the path difference is λ, is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{3}$ will be

(A) $\frac{K}{4}$ (B) K (C) $2K$ (D) $\frac{K}{2}$

Open solution

Q2. NEET 2026

In interference and diffraction, the light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing a bright fringe.

A. As there is no gain or loss of energy, these phenomena are consistent with the principle of conservation of energy.

B. Diffraction and interference are characteristics exhibited only by light waves.

Choose the correct answer from the options given below:

(A) A is true and B is also true (B) A is false, but B is true

(C) A is true, but B is false (D) Both A and B are false

Open solution

Q3. NEET 2025

The intensity of transmitted light when a polaroid sheet, placed between two crossed polaroids at 22.5° from the polarization axis of one of the polaroids, is ($I0_0$ is the intensity of polarised light after passing through the first polaroid):

(A) $\frac{I_0}{8}$ (B) $\frac{I_0}{16}$ (C) $\frac{I_0}{2}$ (D) $\frac{I_0}{4}$

Open solution

Q4. NEET 2025

An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then

(A) Both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to 60° and 30°, respectively

(B) Transmitted light is completely polarized with angle of refraction close to 30°

(C) Reflected light is completely polarized and the angle of reflection is close to 60°

(D) Reflected light is partially polarized and the angle of reflection is close to 30°

Open solution

Q5. NEET 2024 (Re-Examination)

Two slits in Young's double slit experiment are $1.5\ \text{mm}$ apart and the screen is placed at a distance of $1\ \text{m}$ from the slits. If the wavelength of light used is $600 \times 10^{-9}\ \text{m}$ then the fringe separation is

(A) $4 \times 10^{-5}\ \text{m}$ (B) $9 \times 10^{-8}\ \text{m}$ (C) $4 \times 10^{-7}\ \text{m}$ (D) $4 \times 10^{-4}\ \text{m}$

Open solution

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Q6. NEET 2024 (Re-Examination)

Open solution

Interference pattern can be observed due to superposition of the following waves:

A. $y = a \sin \omega t$

B. $y = a \sin 2\omega t$

C. $y = a \sin(\omega t - \phi)$

D. $y = a \sin 3\omega t$

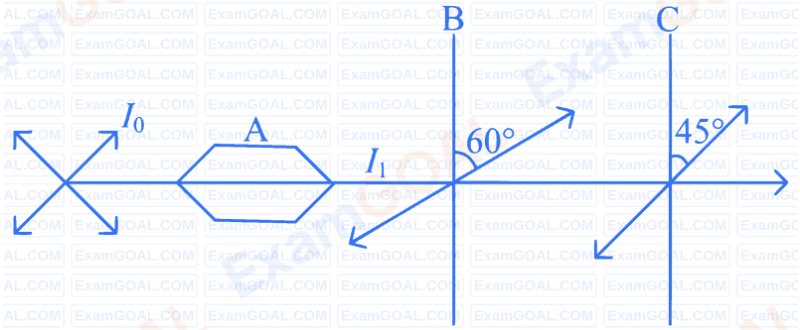
Choose the correct answer from the options given below.

(A) B and C (B) B and D (C) A and C (D) A and B

Q7. NEET 2024 (Re-Examination)

Open solution

A beam of unpolarized light of intensity I0 is passed through a polaroid A, then through another polaroid B, oriented at 60° and finally through another polaroid C, oriented at 45° relative to B as shown. The intensity of emergent light is:



(A) $\frac{I_0}{16}$ (B) $\frac{I_0}{4}$ (C) $\frac{I_0}{2}$ (D) $\frac{I_0}{32}$

Q8. NEET 2024

Open solution

If the monochromatic source in Young's double slit experiment is replaced by white light, then

(A) Interference pattern will disappear

(B) There will be a central dark fringe surrounded by a few coloured fringes

(C) There will be a central bright white fringe surrounded by a few coloured fringes

(D) All bright fringes will be of equal width

Q9. NEET 2024

Open solution

An unpolarised light beam strikes a glass surface at Brewster's angle. Then

(A) The reflected light will be partially polarised.

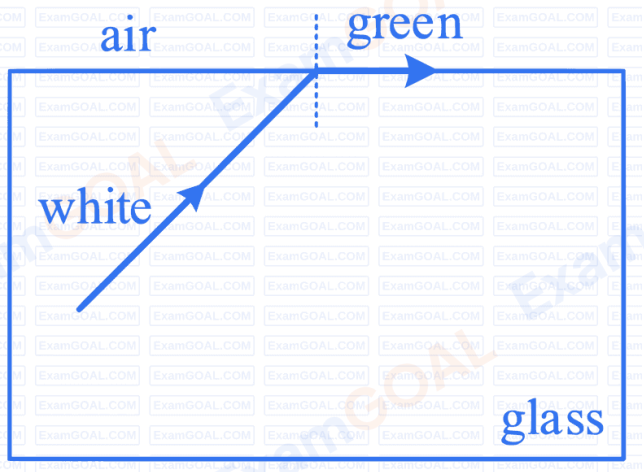
(B) The refracted light will be completely polarised.

(C) Both the reflected and refracted light will be completely polarised.

(D) The reflected light will be completely polarised but the refracted light will be partially polarised.

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Q10. NEET 2023 Manipur



Open solution

Which set of colours will come out in air for a situation shown in figure?

(A) Yellow, Orange and Red
(B) All
(C) Orange, Red and Violet
(D) Blue, Green and Yellow

Q11. NEET 2023

Open solution

For Young's double slit experiment, two statements are given below :

Statement I : If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II : If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false
(B) Statement I is true but Statement II is false
(C) Statement I is false but Statement II is true
(D) Both Statement I and Statement II are true

Q12. NEET 2022 Phase 2

Open solution

If the screen is moved away from the plane of the slits in a Young's double slit experiment, then the :

(A) linear separation of the fringes decreases
(B) angular separation of the fringes increases
(C) angular separation of the fringes decreases
(D) linear separation of the fringes increase

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Q13. NEET 2022 Phase 2

After passing through a polariser a linearly polarised light of intensity I is incident on an analyser making an angle of 30° with that of the polariser. The intensity of light emitted from the analyser will be

(A) $\frac{2I}{3}$ (B) $\frac{I}{2}$ (C) $\frac{I}{3}$ (D) $\frac{3I}{4}$

Open solution

Q14. NEET 2022 Phase 1

In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is

(A) 6 (B) 8 (C) 9 (D) 12

Open solution

Q15. NEET 2020 Phase 1

Assume that light of wavelength 600 nm is coming from a star. The limit of resolution of telescope whose objective has a diameter of 2 m is :

(A) 1.83×10^{-7} rad (B) 7.32×10^{-7} rad

(C) 6.00×10^{-7} rad (D) 3.66×10^{-7} rad

Open solution

Q16. NEET 2020 Phase 1

In Young's double slit experiment, if the separation between coherent sources is halved and the distance of the screen from the coherent sources is doubled, then the fringe width becomes :

(A) half (B) four times (C) one-fourth (D) double

Open solution

Q17. NEET 2019

In a double slit experiment, when light of wavelength 400 nm was used, the angular width of the first minima formed on a screen placed 1m away, was found to be 0.2^o. What will be the angular width of the first minima, (μ_{water} = 4/3) if the entire experimental apparatus is immersed in water?

(A) 0.15^o (B) 0.05^o (C) 0.1^o (D) 0.266^o

Open solution

Q18. NEET 2018

Unpolarised light is incident from air on a plane surface of a material of refractive index μ. At a particular angle of incidence i, it is found that the reflected and refracted rays are perpendicular to each other. Which of the following options is correct for this situation?

(A) Reflected light is polarised with its electric vector parallel to the plane of incidence

(B) Reflected light is polarised with its electric vector perpendicular to the plane of incidence

(C) $i = \sin^{-1}\left(\frac{1}{\mu}\right)$

(D) $i = \tan^{-1}\left(\frac{1}{\mu}\right)$

Open solution

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Q19. NEET 2018

In Young's double slit experiment the separation d between the slits is 2 mm, the wavelength λ of the light used is 5896 $\overset{0}{A}$ and distance D between the screen and slits is 100 cm. It is found that the angular width of the fringes is 0.20^o. To increase the fringe angular width to 0.21^o (with same λ and D) the separation between the slits needs to be changed to

(A) 1.9 mm (B) 1.8 mm (C) 2.1 mm (D) 1.7 mm

Open solution

Q20. NEET 2017

The ratio of resolving powers of an optical microscope for two wavelength λ_1 = 4000 $\overset{\circ}{A}$ and λ_2 = 6000 $\overset{\circ}{A}$ is

(A) 9 : 4 (B) 3 : 2 (C) 16 : 81 (D) 8 : 27

Open solution

Q21. NEET 2017

Young's double slit experiment is first performed in air and then in a medium other than air. It is found that 8th bright fringe in the medium lies where 5th dark fringe lies in air. The refractive index of the medium is nearly

(A) 1.59 (B) 1.69 (C) 1.78 (D) 1.25

Open solution

Q22. NEET 2017

Two polaroids P_1 and P_2 are placed with their axis perpendicular to each other. Unpolarised light I_0 is incident on P_1. A third polaroid P_3 is kept in between P_1 and P_2 such that its axis makes an angle 45° with that of P_1. The intensity of transmitted light through P_2 is

(A) $\frac{I_0}{4}$ (B) $\frac{I_0}{8}$ (C) $\frac{I_0}{16}$ (D) $\frac{I_0}{2}$

Open solution

Q23. NEET 2016 Phase 2

The interference pattern is obtained with two coherent light sources of intensity ratio n. In the interference pattern, the ratio $\frac{I_{max}-I_{\min}}{I_{max}+I_{min}}$ will be

(A) $\frac{\sqrt{n}}{n+1}$ (B) $\frac{2\sqrt{n}}{n+1}$

(C) $\frac{\sqrt{n}}{(n+1)^2}$ (D) $\frac{2\sqrt{n}}{(n+1)^2}$

Open solution

Q24. NEET 2016 Phase 2

A linear aperture whose width is 0.02 cm is placed immediately in front of a lens of focal length 60 cm. The aparture is illuminated normally by a parallel beam of wavelength 5×10^{-5} cm. The distance of the first dark band of the diffraction pattern from the centre of the screen is

(A) 0.10 cm (B) 0.25 cm (C) 0.20 cm (D) 0.15 cm

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Q25. NEET 2016 Phase 1

In a diffraction pattern due to a single slit of width a, the first minimum is observed at an angle 30° when light of wavelength 5000 $\mathring{A}$ is incident on the slit. The first secondary maximum is observed at an angle of

Open solution

(A) $\sin^{-1}\left(\frac{1}{2}\right)$ (B) $\sin^{-1}\left(\frac{3}{4}\right)$
(C) $\sin^{-1}\left(\frac{1}{4}\right)$ (D) $\sin^{-1}\left(\frac{2}{3}\right)$

Q26. NEET 2016 Phase 1

The intensity at the maximum in a Young's double slit experiment is I_0. Distance between two slits is d = 5λ, where λ is the wavelength of light used in the expreriment. What will be the intensity in front of one of the slits on the screen placed at a distance D = 10d ?

Open solution

(A) $\frac{3}{4}I_0$ (B) $\frac{I_0}{2}$ (C) I_0 (D) $\frac{I_0}{4}$

Q27. AIPMT 2015

At the first minimum adjacent to the central maximum of a single-slit diffraction pattern, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the midpoint of the slit is

Open solution

(A) π radian (B) $\frac{\pi}{8}$ radian (C) $\frac{\pi}{4}$ radian (D) $\frac{\pi}{2}$ radian

Q28. AIPMT 2015

Two slits in Young's experiment have widths in the ratio 1 : 25. The ratio of intensity at the maxima and minima in the interference pattern, $\frac{I_{max}}{I_{min}}$ is

Open solution

(A) $\frac{49}{121}$ (B) $\frac{4}{9}$ (C) $\frac{9}{4}$ (D) $\frac{121}{49}$

Q29. AIPMT 2015 Cancelled Paper

For a parallel beam of monochromatic light of wavelength 'λ' , diffraction is produced by a single slit whose width 'a' is of the order of the wavelength of the light. If 'D' is the distance of the screen from the slit, the width of the central maxima will be

Open solution

(A) $\frac{Da}{\lambda}$ (B) $\frac{2Da}{\lambda}$ (C) $\frac{2D\lambda}{a}$ (D) $\frac{D\lambda}{a}$

Q30. AIPMT 2015 Cancelled Paper

In a double slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of double slit within the central maxima of single slit pattern ?

Open solution

(A) 0.5 mm (B) 0.02 mm (C) 0.2 mm (D) 0.1 mm

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Q31. AIPMT 2014

A beam of light of $\lambda = 600$ nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between first dark fringes on either side of the central bright fringe is

(A) 1.2 cm (B) 1.2 mm (C) 2.4 cm (D) 2.4 mm

Open solution

Q32. AIPMT 2014

In the Young's double slit experiment, the intensity of light at a point on the screen where the path difference λ is K, (λ being the wavelength of light used). The intensity at a point where the path difference is $\lambda/4$ will be

(A) K (B) K/4 (C) K/2 (D) zero

Open solution

Q33. NEET 2013 (Karnataka)

In Young's double slit experiment the distance between the slits and the screen is doubled. The separation between the slits is reduced to half. As a result the fringe width

(A) is halved (B) becomes four times

(C) remains unchanged (D) is doubled

Open solution

Q34. NEET 2013 (Karnataka)

A parallel beam of light of wavelength λ is incident normally on a narrow slit. A diffraction pattern is formed on a screen placed perpendicular to the direction of the incident beam. At the second minimum of the diffraction pattern, the phase difference between the rays coming from the two edges of slit is

(A) 2π (B) 3π (C) 4π (D) $\pi\lambda$

Open solution

Q35. NEET 2013 (Karnataka)

The reddish appearance of the sun at sunrise and sunset is due to

(A) the scattering of light (B) the polarisation of light

(C) the colour of the sun (D) the colour of the sky

Open solution

Q36. NEET 2013

In Young's double slit experiment, the slits are 2 mm apart and are illuminated by photons of two wavelengths λ_1 = 12000 $\mathring{A}$ and λ_2 = 10000 $\mathring{A}$. At what minimum distance from the common central bright fringe on the screen 2 m from the slit will a bright fringe from one interference pattern coincide with a bright fringe from the other ?

(A) 4 mm (B) 3 m (C) 8 mm (D) 6 mm

Open solution

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Q37. AIPMT 2007

The frequency of a light wave in a material is 2×10^{14} Hz and wavelength is 5000 $\mathring{A}$. The refractive index of material will be

(A) 1.50 (B) 3.00 (C) 1.33 (D) 1.40

Open solution

Q38. AIPMT 2005

The angular resolution of a 10 cm diameter telescope at a wavelength of 5000 $\mathring{A}$ is of the order of

(A) 10^6 rad (B) 10^{-2} rad (C) 10^{-4} rad (D) 10^{-6} rad.

Open solution

Q39. AIPMT 2004

A telescope has an objective lens of 10 cm diameter and is situated at a distance of one kilometer from two objects. The minimum distance between these two objects, which can be resolved by the telescope, when the mean wavelength of light is 5000 $\mathring{A}$, is of the order of

(A) 0.5 m (B) 5 m (C) 5 mm (D) 5 cm

Open solution

Q40. AIPMT 2001

A ray of light travelling in air has wavelength λ, frequency n, velocity v and intensity I. If this ray enters into water then these parameters are λ', n', v' and I' respectively. Which relation is correct from following ?

(A) $\lambda = \lambda'$ (B) n = n' (C) v = v' (D) $I = I'$.

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Answer Key

1. **A**	2. **C**	3. **A**	4. **C**	5. **D**
6. **C**	7. **A**	8. **C**	9. **D**	10. **A**
11. **B**	12. **D**	13. **D**	14. **D**	15. **D**
16. **B**	17. **A**	18. **B**	19. **A**	20. **B**
21. **C**	22. **B**	23. **B**	24. **D**	25. **B**
26. **B**	27. **A**	28. **C**	29. **C**	30. **C**
31. **D**	32. **C**	33. **B**	34. **C**	35. **A**
36. **D**	37. **B**	38. **C**	39. **C**	40. **B**

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